ML Challenge 2025: Smart Product Pricing Solution Template

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1. Executive Summary

Our approach leverages a **multimodal deep learning model** combining **large language model text features** and **vision transformer image features** for robust product price prediction. Key innovations include **joint feature fusion**, extensive **feature engineering**, and a custom loss function optimizing for real-world pricing accuracy.

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2. Methodology Overview

2.1 Problem Analysis

We interpreted the pricing challenge as a regression problem under strict data-use constraints, requiring only the provided catalog content and product images.
Exploratory Data Analysis (EDA) revealed:

- Product prices are **highly skewed** with notable outliers.
- Many items had descriptive text indicating quantity or packaging, crucial for accurate predictions.
- Image availability varied, requiring robust handling of missing or low-quality visuals.

Key Observations:

- Prices often correlate with quantitative indicators (packs, pieces, counts).
- Some products lack images, demanding fallback feature reliance on text.

2.2 Solution Strategy

We designed a **hybrid multimodal regression pipeline** that fuses deep NLP (text) and computer vision (image) features with engineered numeric features.

Approach Type: Single Model (Multimodal Regression: MMRegressor)
Core Innovation: Joint text-image feature fusion backed by domain-specific engineered features and ensemble-targeted loss function.

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3. Model Architecture

3.1 Architecture Overview

Our model contains **three main branches**:

- Text encoder: Microsoft DeBERTa-v3-large (Huggingface Transformers)
- Image encoder: ViT Base Patch16x224 (timm)
- Numeric feature branch: Standardized text-derived statistics

The outputs are concatenated, normalized, and fed to fully-connected layers for price prediction.

Architecture Diagram:

text

[CLEANED TEXT] → [DeBERTa-v3-large] →

↘

[IMAGE] → [ViT Patch16x224] → [Feature Fusion] → [MLP Regression] → [Price]

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[Engineered Numeric Features] →

3.2 Model Components

Text Processing Pipeline:

- Preprocessing steps: Lowercasing, HTML/unicode normalization, regex cleaning (removing non-alphabetic symbols), missing text handling.
- Model type: DeBERTa-v3-large transformer encoder
- Key parameters: Max length 512 tokens, mean pooling on hidden states

Image Processing Pipeline:

- Preprocessing steps: Prefetching, resizing/cropping to 224x224, RGB normalization, missing image fallback.
- Model type: ViT Base Patch16x224, pretrained on ImageNet
- Key parameters: Feature dimension aggregation via global average pooling

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4. Model Performance

4.1 Validation Results

- **SMAPE Score:** [best validation SMAPE, e.g., 44.55]

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5. Conclusion

Our multimodal approach blends powerful pre-trained NLP and vision encoders with tailored numeric features, yielding a robust price prediction pipeline. The careful exclusion of any external price lookup guarantees fairness and reproducibility. Our feature fusion strategy and ensemble-aware loss function ensure strong real-world accuracy, even on outlier products.
